Article

The Bell Boundary Condition: Retrocausal Worldtube Interpretation of a Deterministic Brane-Lattice Substrate (Paper V)

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Abstract

Bell's theorem rules out theories that are simultaneously deterministic (D), local (L), and measurement-independent (MI). The deterministic brane-lattice substrate of Paper I satisfies (D) by construction and (L) by its nearest-neighbor spring dynamics; (MI) must therefore be relaxed. We argue that, within the specific ontology of this substrate, the retrocausal loophole is not merely *a* consistent option but the *unique* consistent completion: superdeterminism is ruled out because the local elastic medium has no mechanism to coordinate microscopic initial conditions with macroscopic measurement choices at spacelike separation, and nonlocal hidden-variable completions require adding superluminal influence channels absent from the spring-lattice dynamics. The retrocausal worldtube reading follows directly from the time-symmetry of the brane action, which selects a *kind* of direction (timelike vs spacelike) but not a *direction* along the timelike axis. The arrow of time emerges at the soliton level: matter solitons are forward-propagating worldtubes; antimatter solitons are backward-propagating worldtubes, realizing the Feynman–Stueckelberg interpretation at the level of substrate solutions. An entangled pair is one branching 4D world-volume splitting at a V-vertex in the shared past; correlations propagate backward along one branch to the V-vertex and forward along the other, with no spacelike signal at any step. The cosmological matter–antimatter asymmetry is a geometric initial-condition asymmetry: the two chirality branches emerge from the Big Bang vertex in opposite time directions, and only the forward branch is observable. Three open simulation tests are identified: chirality classification, V-vertex correlation, and Born-rule recovery.

Keywords: Bell's theorem; retrocausality; worldtube; soliton chirality; time-symmetric dynamics; deterministic substrate; brane lattice; measurement independence; V-branching; entanglement

1. Introduction

This is the fifth paper in the series. Paper I established the substrate model and stated the Bell interpretive commitment as a brief section within the core paper. The present paper separates that commitment into a standalone treatment, develops the uniqueness argument for the retrocausal reading, and traces its consequences for how the simulation program should interpret its own results.

Why Bell cannot be left implicit. The substrate of Paper I is deterministic (Equation (26) of Paper I) and local (all forces are nearest-neighbor spring contacts at finite propagation speed). Bell's theorem then forces an interpretive stance before any specific bridge is developed: it rules out a large class of classical-substrate completions and constrains which

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interpretations are available. Leaving the Bell discussion buried inside the core paper risks treating it as an optional gloss. A standalone paper makes explicit that the retrocausal worldtube reading is not a stylistic preference but a *forced consequence* of the substrate's own ontological commitments.

The uniqueness argument. Bell's theorem rules out theories that are simultaneously (D) deterministic, (MI) measurement-independent, and (L) local in the sense of finite-speed influence propagation [1,2]. The substrate satisfies (D) by construction and (L) by its nearest-neighbor dynamics. This means (MI) must be relaxed. Two ways to relax (MI) are available in principle:

- *Superdeterminism:* correlate detector settings with the hidden state at some common past boundary condition. This is logically consistent but explanatorily circular: it assumes the very correlations it is supposed to explain [3].
- *Retrocausality:* allow the future measurement context to propagate *backward* along the worldtube to the shared past of the entangled pair.

The substrate's own ontology closes the superdeterminism route. Superdeterminism requires a cosmic conspiracy between detector choices and the initial substrate configuration; but the substrate is a local elastic medium, not a pre-programmed Oracle. It has no mechanism to coordinate microscopic initial conditions at spacelike separation with macroscopic measurement choices made later. The retrocausal reading, by contrast, requires only what the 4D brane ontology already provides: a 4D-in-4D world-volume in which future boundary conditions are as natural as past ones. Under (D) + (L) + rejection of superdeterminism, retrocausality is the *unique* consistent completion available to this substrate.

Paper organization. Section 2 lists imports from Paper I. Section 3 develops the Bell constraint and the uniqueness argument in detail. Section 4 develops the retrocausal worldtube interpretation: time-symmetry of the brane action, causality from soliton chirality, matter and antimatter as opposite-chirality worldtubes, entanglement as V-branching, and implications for the simulation program.

2. Interfaces

This paper imports the following results (chiefly from Paper I, plus the localized-mode result of Paper IV) without re-derivation. As the interpretive capstone of the series it has no downstream dependents and exports no interfaces.

Substrate ontology. The 4D cubic brane lattice embedded codimension-zero in a 4D Euclidean ambient, with the timelike direction selected by the Lorentzian sign structure of the brane action (Section 3.1). The substrate is manifestly local: all forces propagate through direct spring-neighbor contact at finite speed.

Time-symmetry of the brane action. The action (Equation (24)) is time-symmetric: it selects a *kind* of direction (timelike vs spacelike) but not a *direction* along the timelike axis (past-to-future vs future-to-past). The arrow of time is therefore not a property of the substrate.

Soliton chirality. The concept that localized modes (Paper IV Section 3) can propagate forward or backward along the brane's timelike axis, selecting the causal direction at the soliton level rather than at the substrate level.

Determinism. The substrate equations of motion (Equation (26)) are deterministic: given a complete initial configuration, all future and past configurations are uniquely fixed.

3. The Bell constraint and the uniqueness argument

3.1. Bell's theorem and its assumptions

Bell's theorem rules out theories that are simultaneously:

(D) *Deterministic*: hidden variables λ fully fix outcomes given the measurement settings a, b.

(L) *Local*: the outcome at one wing depends only on the local setting and λ, not on the setting at the distant wing.

(MI) *Measurement-independent*: the joint distribution of hidden variables and detector settings factorizes, $P(\lambda, a, b) = P(\lambda)P(a, b)$, so the hidden state is statistically independent of the measurement choices.

Any theory satisfying all three reproduces only correlations bounded by the Bell inequalities [1,2]. The experimental violation of Bell inequalities — now with all standard loopholes closed [4–6] — therefore implies that at least one of (D), (L), (MI) must be given up by any classical-substrate completion.

3.2. Which assumption the substrate can give up

The substrate of Paper I satisfies (D) and (L) by construction:

- **(D) is non-negotiable.** The equations of motion (Equation (26) of Paper I) are second-order hyperbolic PDEs with deterministic initial-value evolution. Any probabilistic appearance at the macroscopic level must be emergent (Paper IV Section 4.1), not fundamental. Giving up (D) would discard the substrate hypothesis itself.
- **(L) is non-negotiable.** The substrate forces are nearest-neighbor spring contacts. Information propagates at finite speed (c_T or c_L, Paper I Equation (17)). Giving up (L) would require adding a new superluminal interaction channel to the substrate — an entirely different model.

Therefore (MI) must be relaxed.

3.3. Why superdeterminism is ruled out by the substrate ontology

One way to relax (MI) is superdeterminism: the hidden state λ is correlated with detector settings a, b through a common past boundary condition, so $P(\lambda|a, b) \neq P(\lambda)$ even without any causal influence between λ and the measurement choices.

Superdeterminism is logically consistent but is closed off by the substrate's own ontology for two reasons:

No coordination mechanism. The substrate is a local elastic medium with nearest-neighbor dynamics. It has no mechanism to produce microscopic initial correlations between the degrees of freedom that will later constitute detector orientations (macroscopic, made by experimenters) and the hidden-variable state of an entangled pair prepared at an earlier time and place. Such coordination would require either a non-local coupling (violating (L)) or an elaborate fine-tuning of the initial state that is itself unexplained by the substrate dynamics.

Explanatory circularity. Superdeterminism explains observed correlations by positing correlations in initial conditions. The initial conditions are not derived from the substrate dynamics; they are assumed. The substrate hypothesis is precisely an attempt to *derive* structure from minimal commitments; invoking superdeterminism undoes this programme by hiding the correlations in an unobservable past [3].

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3.4. Retrocausality as the unique remaining option

The only remaining way to relax (MI) without adding a coordination mechanism is to allow the future measurement context to influence the past state of the particle along its worldtube. This is the *retrocausal* option:

> $P(\lambda|a,b) \neq P(\lambda)$ because the future boundary condition set by measurement a back-propagates along the worldtube to the emission event, correlating λ with a without any spacelike influence.

This requires only what the 4D brane action already provides: a 4D world-volume in which future boundary conditions are as natural as past ones (Section 2).

Counter to the objection that retrocausality is not uniquely forced. A common objection is that retrocausality is not the *only* option: one could add explicit nonlocal hidden variables (e.g. pilot-wave / Bohmian mechanics) that respect (D) and (MI) while giving up the *local* sense of (L). This is correct in the abstract, but Bohmian nonlocality requires a preferred foliation and a nonlocal guidance equation — structures that conflict with the substrate's local elastic ontology. In the specific context of a locally-interacting 4D brane lattice, the options are:

1. Give up (D): not available (the substrate is deterministic by construction).
2. Give up (L) via a nonlocal hidden-variable mechanism: not available without adding superluminal influence channels beyond the spring-lattice dynamics.
3. Give up (MI) via superdeterminism: ruled out by the substrate's lack of a coordination mechanism, as above.
4. Give up (MI) via retrocausality: available; requires only the 4D worldtube structure already present in the substrate ontology.

Under the specific commitments of this substrate — deterministic, locally-interacting, 4D brane — retrocausality is therefore the unique consistent completion. This is a stronger claim than the general statement that retrocausality is *a* consistent option; it is the claim that it is the *only* option the substrate's own dynamics supports.

3.5. Relation to existing retrocausal interpretations

The retrocausal worldtube reading places itself in the same interpretive family as:

- the two-state-vector formalism of Aharonov and collaborators [7,8], which assigns both a forward-propagating and a backward-propagating state to each quantum system;
- Cramer's transactional interpretation [9], in which a transaction is completed by a retarded offer wave and an advanced confirmation wave;
- the retrocausal models of Price and Wharton [10], which systematically develop the ontological consequences of time-symmetric dynamics;
- Sutherland's time-symmetric Bohmian model [11], which grounds retrocausality in a concrete pilot-wave picture.

The present theory differs from all of these in one structural respect: the retrocausal propagation is not imposed as an interpretive rule on top of a forward-causal dynamics. It is *built into the substrate action*, which is time-symmetric by construction. The retrocausal interpretation is therefore not an add-on; it is the direct reading of what the brane action says.

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4. The retrocausal worldtube interpretation

4.1. Time-symmetry of the substrate action

The brane action of Paper I (Equation (24)) is time-symmetric. Its Lorentzian sign structure selects a *kind* of direction — timelike vs spacelike — but the action is invariant under reversal of the timelike coordinate, $t \mapsto -t$. Neither the kinetic term nor the elastic potential distinguishes past from future. This means the substrate has no intrinsic arrow of time: both forward-in-time and backward-in-time solutions to Equation (26) of Paper I are equally valid.

4.2. Causality from soliton chirality

If the substrate has no intrinsic arrow of time, where does the empirical arrow of time come from?

In the retrocausal worldtube reading, the arrow of time is a property of *solitons*, not of the substrate. A soliton solution to the substrate equations of motion (Equation (26) of Paper I) may propagate forward or backward along the brane's timelike axis. The choice of direction is the *chirality* of the soliton: a topological/dynamical property of the solution analogous to the handedness of a Skyrme texture.

- **Matter solitons** are forward-propagating worldtubes: their chirality selects the future direction as the direction of propagation.
- **Antimatter solitons** are backward-propagating worldtubes: their chirality selects the past direction.

This realizes the Feynman–Stueckelberg interpretation of antiparticles as time-reversed particles [12,13] at the level of substrate soliton solutions, rather than as an interpretive remark attached to a Lorentz-invariant field theory after the fact.

Cosmological asymmetry. The empirical excess of matter over antimatter is, on this reading, a *geometric initial-condition asymmetry*: matter and antimatter solitons emerge from the Big Bang vertex propagating in opposite time directions along the brane's selected timelike axis, and our observable universe is the forward-propagating branch. This differs structurally from Sakharov-condition baryogenesis [14], which requires the same time direction to host both matter and antimatter with a small dynamical asymmetry favouring matter. Here, the asymmetry is not dynamical but geometric: the two branches simply do not share the same observable volume.

4.3. Entanglement as V-branching worldtubes

An entangled pair of solitons, on the present reading, is *one* branching 4D world-volume whose worldtube splits at a V-vertex in the shared past.

Mechanism. Each branch of the world-volume propagates forward to a measurement event. The boundary condition imposed by a measurement on one branch is not transmitted *sideways* through space to the other branch (which would be a spacelike influence, violating (L)). Instead, it propagates *backward* along the branch's own worldtube to the V-vertex in the shared past, and from there *forward* along the other branch to the distant measurement. The V-vertex is the mediator; no spacelike signal is required.

Locality along each worldline. The correlation is:

- *local along each worldline*: no influence crosses a spacelike boundary at any step of the propagation.
- *deterministic given the full 4D configuration*: the correlation is fixed by the worldtube geometry and the V-vertex boundary condition.

- *Bell-inequality-violating*: the joint measurement context — both settings a and b — affects the past of both branches via the V-vertex, so the standard factorizability assumption of Bell's derivation fails.

The 4D world-volume picture promotes "spooky action at a distance" to "ordinary continuity of one extended elastic object."

No preferred foliation required. Unlike Bohmian mechanics, this picture does not require a preferred spacelike foliation. The correlation propagates along the worldtube in a covariant, 4D-in-4D sense. The effective Lorentz symmetry of Paper II constrains which foliations are observationally available to inside observers, but the worldtube itself is a 4D geometric object prior to any foliation choice.

4.4. Implications for the simulation program

The retrocausal worldtube reading reshapes how the simulation program of Papers II–IV should interpret its own results.

Eigenstates as worldtubes. A converged periodic-BC eigenstate (Paper IV Section 3.6) is not a static snapshot; it is a closed 4D worldtube on which the substrate field satisfies the time-symmetric equations of motion with a periodic closure condition. The carrier frequency ω_0 fixed by the closure condition $\omega T = 2\pi n$ is the timelike winding number of the worldtube, not a clock reading.

Entanglement test. The V-branching picture makes a concrete simulation prediction: two solitons initialized with correlated phases at a common point in spacetime should maintain those correlations over spacelike separation in a way that — when the full 4D worldtube is inspected — is traceable to continuity through the V-vertex, not to any spacelike link. A simulation that finds correlation without spacelike signal is consistent with the retrocausal reading; one that requires adding a nonlocal coupling is not.

Matter vs antimatter. Two soliton solutions with opposite chirality (forward vs backward propagating) should appear as particle–antiparticle pairs: identical mass and spin, opposite quantum numbers, and opposite time orientation of the closed worldtube. A simulation that finds such paired solutions would directly confirm the soliton-chirality mechanism.

4.5. Open problems

1. **Explicit chirality classification.** Define the soliton chirality quantum number precisely in terms of the topological winding (Paper IV Equation (4)) and the time orientation of the worldtube closure condition. Show that two solutions related by $t \mapsto -t$ carry opposite chirality.
2. **V-vertex simulation.** Simulate two solitons initialized at a common point with opposite time orientations; verify that the subsequent 4D world-volumes share a V-vertex and exhibit correlated behavior at their respective measurement endpoints without any spacelike signal.
3. **Born-rule recovery.** Identify the substrate mechanism that produces apparent Born-rule statistics from the deterministic worldtube dynamics. The candidate mechanism (Paper IV Section 4.1): Fourier-localization constraints on wavepacket overlaps combined with nonlinear exchange events at the V-vertex. A quantitative model is not yet available.

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Appendix A. Symbol dictionary

t	Slicing parameter along the brane's timelike direction; not an external/ambient time.
α	Dimensionless pre-stress parameter of the substrate (Paper I).
c_L	Long-wavelength longitudinal wave speed (Paper I Equation (17)).
c_T	Long-wavelength transverse wave speed (Paper I Equation (17)).

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